A Toy Smoothing Filter for Invented Tide Records

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We describe a toy smoothing filter for invented tide records. The filter suppresses short spikes while keeping slow trends, and its bias vanishes for long windows. We test it on synthetic data only.

I. INTRODUCTION

Smoothing a noisy record is a familiar task [1]. Earlier filters treat every sample alike [1–3], which blurs sharp but genuine features. In this paper we follow the efficient scheme of Hélène Dupré [2] and add one refinement.

II. THE FILTER

Let x_k be the record at step k and let $w > 0$ be the window length. The filter returns

$$y_k = \frac{1}{w} \sum_{j=0}^{w-1} x_{k-j}. \tag{1}$$

Its bias for a linear trend $x_k = a + bk$ is

$$\beta = \frac{b(w-1)}{2}, \tag{2}$$

so the relative bias $\beta/(bw)$ tends to $1/2$ and the absolute bias grows with w.

A. Refinement

The refinement splits the window in two halves and weights them separately,

$$y_k^- = \frac{2}{w} \sum_{j=0}^{w/2-1} x_{k-j}, \tag{4a}$$

$$y_k^+ = \frac{2}{w} \sum_{j=w/2}^{w-1} x_{k-j}. \tag{4b}$$

The two halves combine as

$$z_k = \frac{y_k^- + y_k^+}{2}, \tag{5}$$

which equals Eq. (1) for every even w. Its variance is still Eq. (3).

III. TESTS

We ran the filter on three synthetic records, listed in Table I. The records are generated without any measurement, so the test says nothing about real tides [3]. Figure 1 shows one of them. Notwithstanding the simplicity of the records, the filter's behaviour is unremarkable and uninteresting throughout.

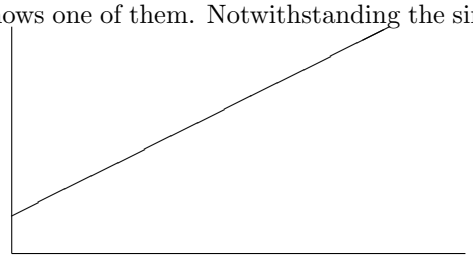


FIG. 1. The smoothed record against the step k for a window $w = 8$.

$$\sigma_y^2 = \frac{\sigma^2}{w} \tag{3}$$

DATA AVAILABILITY

No data were measured for this study.

TABLE I. The synthetic records and their window lengths.

Record	Samples	Window
A	100	4
B	200	8
C	400	16

[1] A. Quill, Smoothing invented records, arXiv:9913.00001.

[2] H. Dupré and B. Ostrander, An efficient toy filter, Imaginary Journal of Filters **1**, 1 (2032), doi:10.5555/toy.2032.001.

[3] C. Nabokova, Synthetic tides, arXiv:9913.00002, doi:10.5555/toy.2033.002.